UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 005 · Particular integral Type 3 (X = x^m) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
added for clarity	Standard context that is not in the class material
transcript only	Said in the lecture but not on any slide
video only	Seen only in a frame of the class video
not on current slides	From older material or past papers, not this year's slides
S005·8	Unit 2 deck 005 (“Particular integral of Type 3”), slide 8 (rendered .pptx)
T005 00:06:48	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 3” (12.6 min), segment start
T006a 00:06:05	Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min), its Type 3 part 00:00:00–00:09:55; the class 006 pack cites the whole file as T006b
CW U2 P15	Unit 2 class-work problem 15
ESA Jul-23 3b	Past paper: UE22MA141A Jul 2023, question 3.b
ESA Jan-24 s2 2d	Past paper: UE23MA141A Jan 2024 set 2, question 2.d

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3. Figure G3.1 · A factor D raises the degree of the PI
4. Figure G4.1 · Superposition in ESA Jul-23 3b: three PIs added

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = the particular integral or the right-hand side, vermilion = a second curve (other solutions, the PI when D is a factor), green = a third (the polynomial part); grey is used for terms that vanish. Every plotted PI was checked by substituting it back with sympy.

1 The Type 3 rule: $X = x^m$

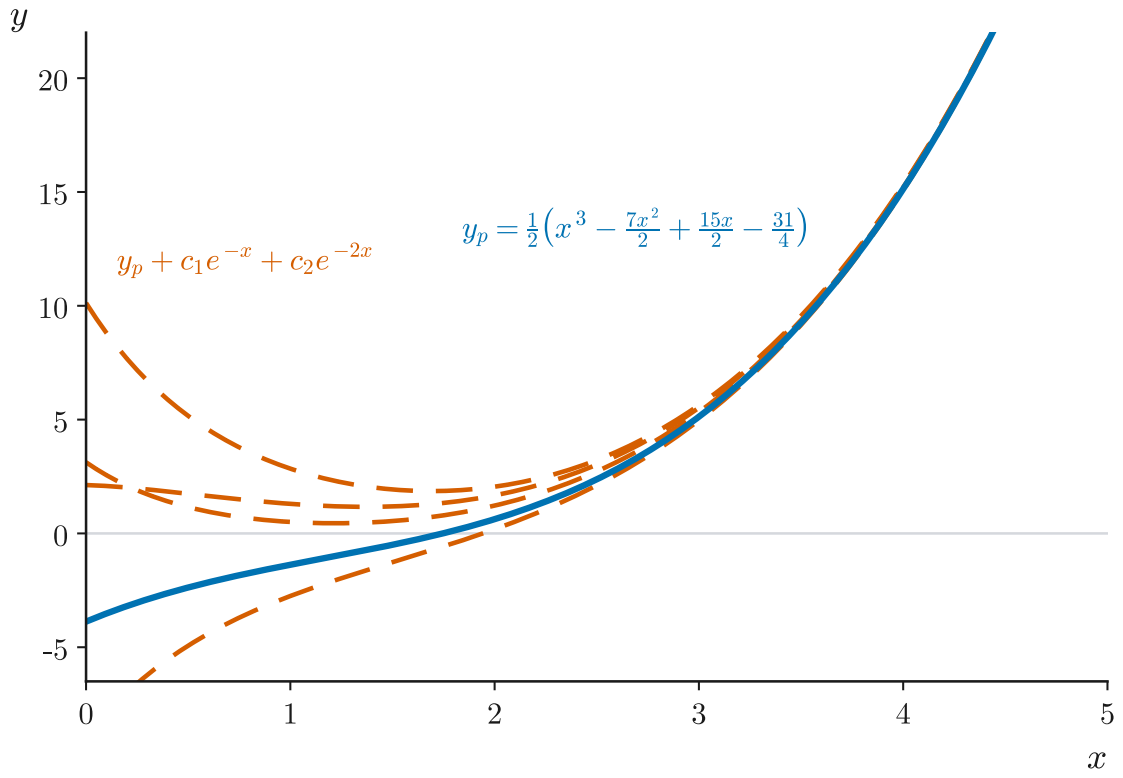


Figure G1.1. Every solution of $(D^2 + 3D + 2)y = x^3 + x^2$ settles onto the PI. ESA ★★

ADDED FOR CLARITY S005·11 · S005·12

How to read it. The PI $y_p = \frac{1}{2}(x^3 - \frac{7x^2}{2} + \frac{15x}{2} - \frac{31}{4})$ (blue) is a cubic, the same degree as X. Four general solutions $y_p + c_1e^{-x} + c_2e^{-2x}$ (vermilion dashed) start at different values but the CF terms die out (roots -1, -2), so they all merge with y_p within a few units of x.

2 The binomial expansions and where to stop

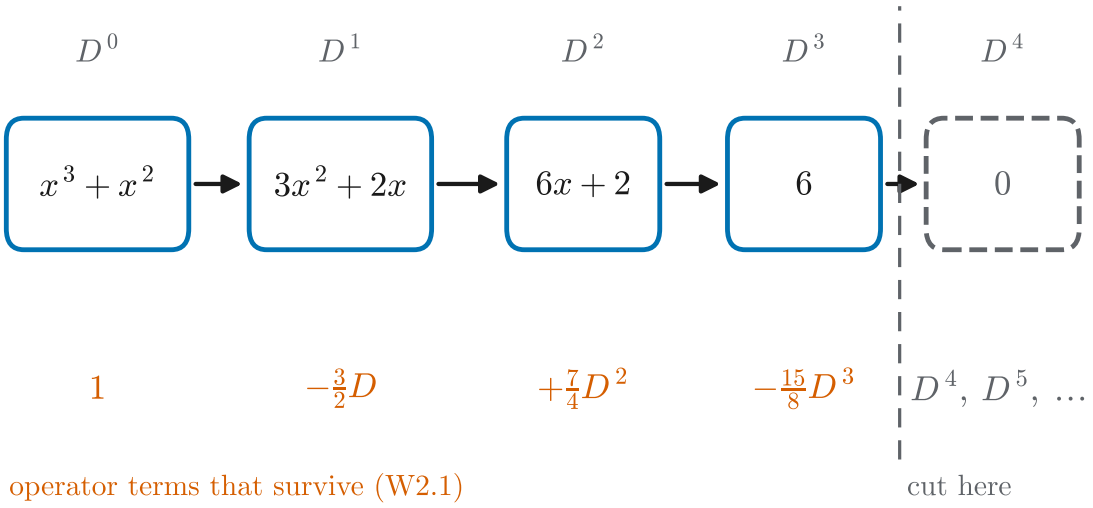


Figure G2.1. Why the series stops at D^m: the derivative staircase of $x^3 + x^2$. ESA ★★★

ADDED FOR CLARITY S005·4 · S005·10 · S005·12 · T006a 00:05:45

How to read it. Each step to the right is one more D. After three steps the polynomial is the constant 6, and D^4 gives 0 (grey). So in $[1 + \frac{D^2+3D}{2}]^{-1}$ only the operator terms $1, D, D^2, D^3$ (shown under the boxes with their coefficients from W2.1) can contribute; D^4, D^5, D^6 are dropped (schematic).

3 When D is a factor of $f(D)$

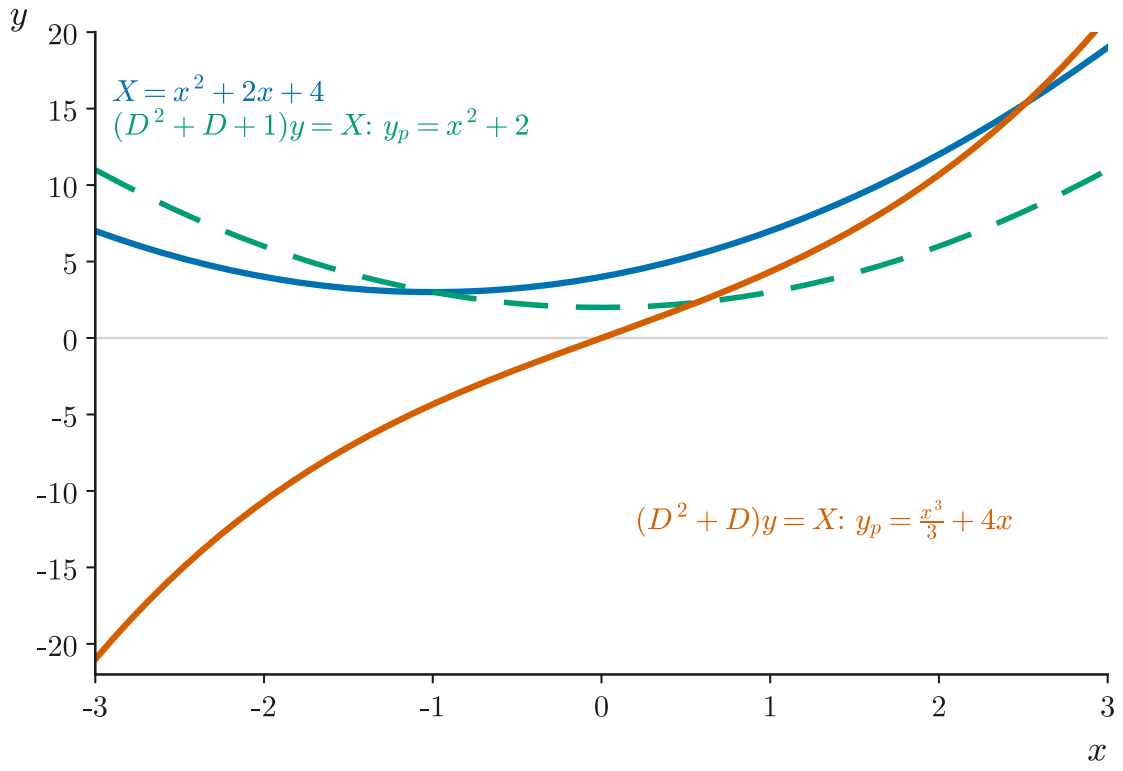


Figure G3.1. A factor D raises the degree of the PI. ESA ★

ADDED FOR CLARITY S005·4

How to read it. Same right-hand side $X = x^2 + 2x + 4$ (blue). With $f(D) = D^2 + D + 1$ ($f(0) = 1$) the PI $x^2 + 2$ (green dashed) is a quadratic like X. With $f(D) = D^2 + D$ (D a factor) the PI $\frac{x^3}{3} + 4x$ (vermilion) is a cubic: the $\frac{1}{D}$ integrates once.

4 Mixed right-hand sides: Types 1, 2 and 3 together

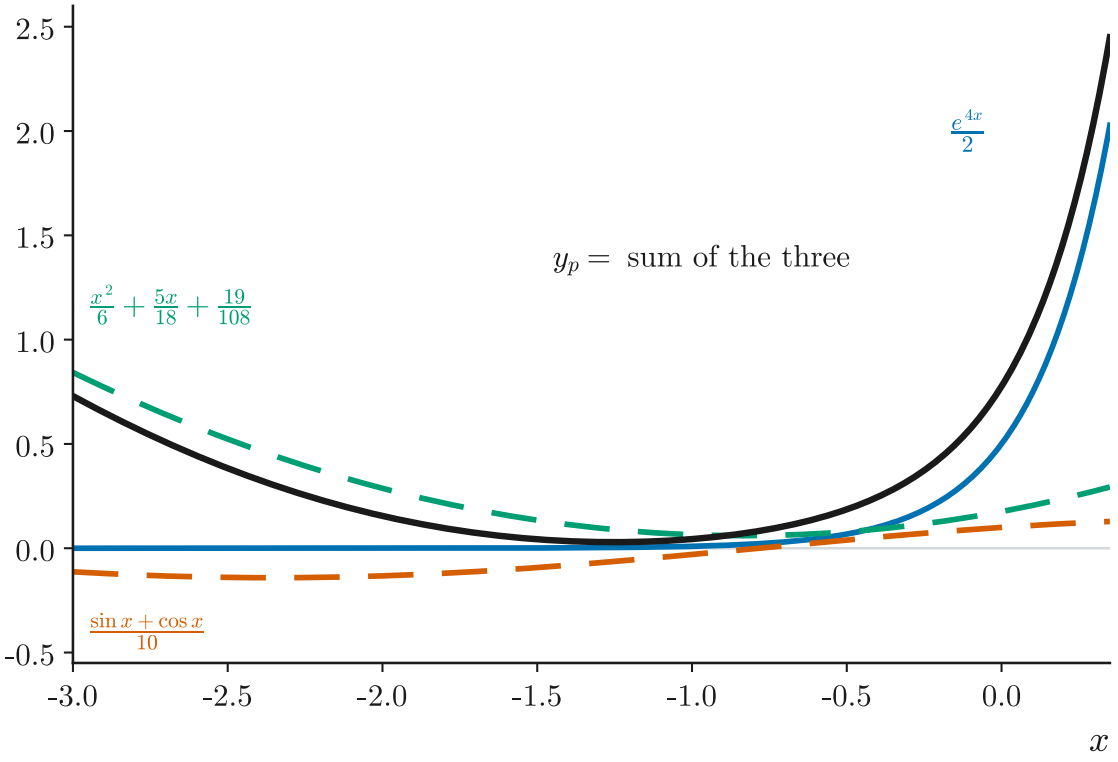


Figure G4.1. Superposition in ESA Jul-23 3b: three PIs added. ESA ★★★

ADDED FOR CLARITY ESA Jul-23 3b

How to read it. The three parts of the PI of $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$: $\frac{e^{4x}}{2}$ (blue, Type 1), $\frac{\sin x + \cos x}{10}$ (vermilion dashed, Type 2), $\frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$ (green dashed, Type 3), and their sum (black). Each part is found separately; the PI is the plain sum.